TECHNICAL DOCUMENTATION

RAG-Based Customer Support Assistant using LangGraph and HITL

Introduction

Retrieval-Augmented Generation (RAG) is an advanced architecture that improves response accuracy by retrieving relevant contextual information from external knowledge sources before generating answers. Traditional chatbots depend only on pretrained knowledge, which often leads to hallucinated responses. The proposed system integrates document retrieval with language model processing to ensure reliable and context-aware responses for customer support applications.

System Architecture Explanation

The system consists of multiple interconnected modules that process user queries through a structured workflow. Initially, the PDF knowledge base is loaded and divided into smaller chunks. These chunks are converted into embeddings and stored in ChromaDB. When a user submits a query, the system retrieves relevant document chunks using similarity search. The retrieved context is processed by the LLM layer to generate responses. If the response confidence is insufficient, the query is escalated to a human agent using the HITL module.

Design Decisions

The chunk size of 500 tokens with 50-token overlap was selected to preserve contextual continuity while maintaining retrieval efficiency. Sentence Transformers were used for embedding generation due to their lightweight architecture and strong semantic understanding capability. ChromaDB was selected as the vector database because it supports efficient similarity search operations and easy local deployment. LangGraph was used to implement workflow orchestration between system nodes.

Workflow Explanation

The workflow begins when the user submits a query through the command-line interface. The query is passed to the LangGraph input node, which initiates workflow execution. The processing node retrieves relevant document chunks from ChromaDB. The decision node evaluates similarity score and response confidence. If confidence is sufficient, the output node generates the response. Otherwise, the system forwards the query to the HITL module for escalation.

Conditional Logic

The system uses conditional routing logic to determine whether an automatic response should be generated or escalation is required. Automatic responses are generated when similarity score exceeds the predefined threshold. Escalation occurs when similarity score is low, context is missing, or the query falls outside the knowledge base.

HITL Implementation

The Human-in-the-Loop module improves system reliability by allowing complex queries to be handled by human agents. When escalation is triggered, the query is forwarded to the agent interface. The agent reviews the request and provides the final response, which is returned to the user through the output node.

Challenges and Trade-offs

Larger chunk sizes improve contextual understanding but increase retrieval latency. Smaller chunk sizes reduce latency but may weaken semantic coverage. High-dimensional embeddings improve retrieval accuracy but increase storage requirements. Selecting optimal chunk size and embedding models ensures a balance between performance and efficiency.

Testing Strategy

The system was tested using multiple customer-support-related queries such as:

- What is the refund policy?
- How can I reset my password?
- When is escalation triggered?
- Connect me to a human agent

The responses confirmed accurate retrieval from the knowledge base and proper escalation handling for unsupported queries.

Future Enhancements

Future improvements include support for multiple document ingestion, integration with web-based interfaces, implementation of conversational memory, deployment using cloud-based vector databases, and addition of feedback-driven learning mechanisms.